EXPERIMENT 17

The Hall Effect

The purpose of this experiment is to demonstrate the Hall effect for an indium arsenide semiconductor. The experiment shows how the Hall effect can be used to determine the concentration of charge carriers in the semiconductor, and also, the sign of the charge carriers.

Theory

If a current-carrying conductor is placed at right angles to a magnetic field, it is found that a transverse potential difference (Hall Voltage) is developed across the sides of the conductor. This is known as the Hall effect which was discovered in 1897 by E.H. Hall.

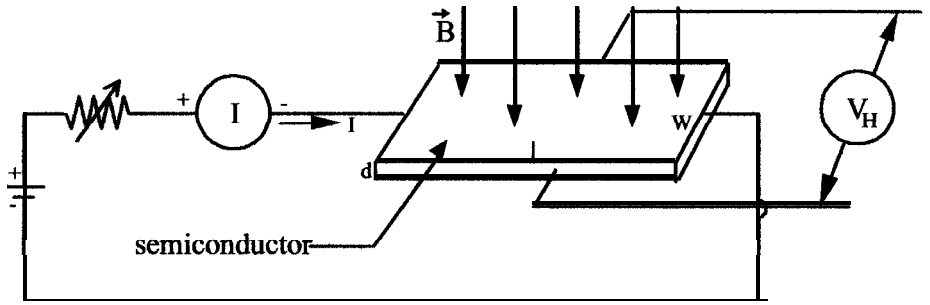


Figure 1

Suppose the conductor consists of a thin rectangular semiconductor carrying a current I as shown in Figure 1. If a magnetic field B is applied perpendicularly to the faces of the slab, a Hall voltage V_H will appear across the width of the slab according to the equation

$$V_H = \frac{R_H B I}{d} \qquad (1)$$

where d is the thickness of the semiconductor and R_H is the Hall coefficient.

The Hall voltage V_H can be theoretically explained by considering the forces acting on the charge carriers inside the slab. For a conventional current I as shown in Figure 1, the charge carriers may consist of positive charges moving to the right, or else, negative charges moving to the left. The Hall effect can be used to decide between these two possibilities.

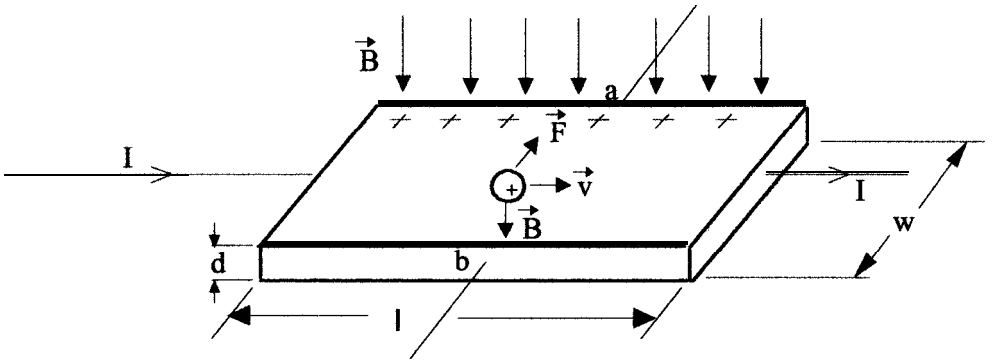


Figure 2

For discussion purposes, suppose the carriers consist of positive charge q moving to the right with a drift velocity v as shown in Figure 2. The applied magnetic field B exerts a magnetic force $\mathbf{F_m}$ on the charge q according to the Lorentz equation

$$\mathbf{F}_m = q\mathbf{v} \times \mathbf{B} \tag{2}$$

Since the vectors V, B, F are mutually perpendicular, the magnitude of the force becomes

$$F_m = qvB \tag{3}$$

As a result of this force, the charges are deflected towards the upper edge of the slab so that there will be an accumulation of charges on the top "a" compared to the bottom "b". The charge accumulation sets up a potential difference V_H and a corresponding electric field E_H given by

$$E_H = \frac{V_H}{w} \tag{4}$$

where w is the width of the slab. The electric field $\mathbf{E}_H$ is opposite in direction to the magnetic force, so that eventually, the electric force $\mathbf{F}_e = q\mathbf{E}_H$ counterbalances the magnetic force $\mathbf{F}_m$ Thus we have

$$F = q\,E_H = qvB$$

or

$$\frac{V_H}{w} = vB \tag{5}$$

The drift velocity v can be found from the current I as follows

$$v = \frac{I}{nqA} \tag{6}$$

where n is the number of charge carriers per unit volume and A = wd is the cross-sectional area.

Substituting equation (6) into equation (5), we obtain the final expression for V_H given by

$$V_H = \frac{1}{nq}\frac{BIw}{A} = \frac{1}{nq}\frac{BI}{d} \tag{7}$$

Comparing equation (7) to equation (1), the Hall coefficient R_H is defined as

$$R_H = 1/nq. \tag{8}$$

Hall effect studies can give valuable information about the electrical conduction process in metals and semiconductors. The sign of the Hall Voltage V_H can be used to show whether electrons or whether positive charges (holes) are the dominant carriers in the material. The Hall coefficient R_H can be used to obtain the number of charge carriers per m^3. If the carriers are electrons of charge -e or holes of charge +e, the concentration n is given by

$$n = 1/R_H e. \tag{9}$$

Indium Arsenide Semiconductor

Unlike metals where the charge carriers are electrons, semiconductors can have both positive and negative carriers. In an intrinsic semiconductor, there are equal numbers of electrons and holes. This only happens in an extremely pure semiconductor where no impurities or defects occur.

Most semiconductors are doped with impurity atoms so as to greatly increase the number of charge carriers. In an n-type semiconductor, the addition of impurity atoms contributes extra electrons so that the majority carriers are said to be electrons. In a p-type semiconductor, the impurity atoms have a shortage of electrons so that holes (vacancies left by electrons) are the dominant factor for the conduction of electric current.

The Hall cell used in this experiment is made from an indium arsenide semiconductor. The cell can be either n-type or p-type depending on how much excess arsenic or indium atoms are added. Indium arsenide was chosen since it has a large Hall coefficient which does not vary much with temperature.

Apparatus

There are two different Hall probes used in this experiment (bell and Cenco). Both probes have indium arsenide as the semiconductor sample. The probes are pre-wired to a "built-in" circuit box which enables adjustment of the current I and measurement of the Hall voltage V_H. The wires are color-coded to identify which leads correspond to the current and which leads correspond to the Hall voltage. The red and black leads are used for the current I, while the yellow and green (or white and green) leads are used for the Hall voltage V_H.

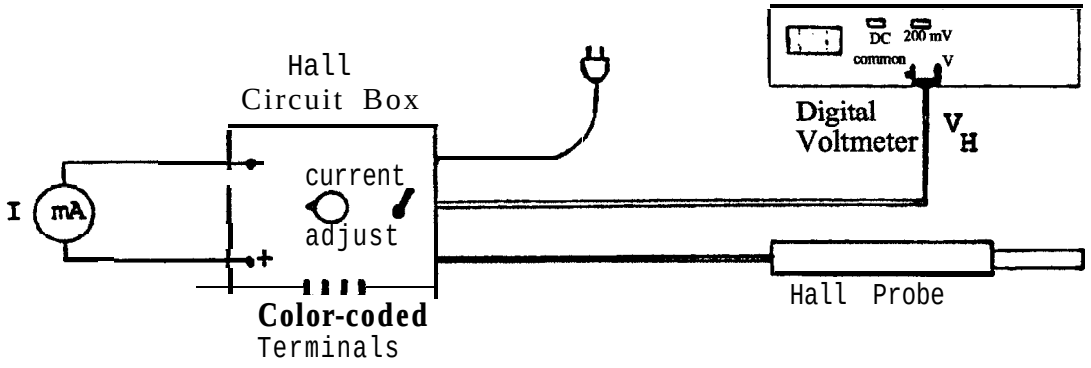
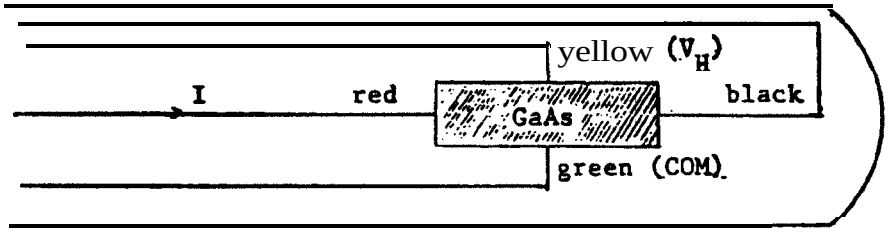


Figure 3

Procedure

Part I - Bell Probe

1. Plug in the Hall circuit box for the Bell Probe and set the current I to 150 mA. Connect the Hall voltage cable (with banana plug) to a digital voltmeter. With no magnetic field (B = 0), measure the residual voltage V_o of the Hall probe. The residual voltage is due to imperfect centering of the Hall electrodes on opposite sides of the semiconductor slab.



d = 0.114 mm
w = 2.03 mm
ℓ = 4.57 mm

Figure 4

2. Determine the sensitivity V_H/B of the Bell probe by placing the probe-end in a known magnetic

field B of a reference magnet (2.91 kilogauss). The magnetic field of this device acts downward ↓. Rotate the probe until a maximum Hall voltage is obtained (Why?), The final voltage should be corrected to account for the residual voltage.

3. Calculate the Hall coefficient R_H and the number of charge carriers per unit volume.

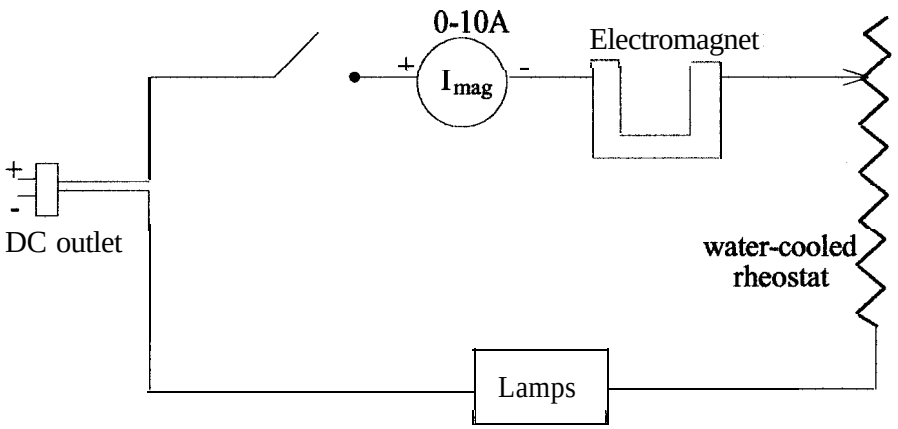


Figure 5

4. Because the Hall voltage is directly proportional to the magnetic field when a constant current I is applied to the probe, it is a very useful device for measuring unknown magnetic fields. Use the calibrated Bell probe to determine the magnetic field B_{mag} of a large electromagnet which is driven by a current I_{mag} as shown in Figure 5. Obtain a compass from the Equipment Room 236 and determine the direction of B_{mag}. Plot a graph of B_{mag} versus I_{mag}. Turn on the water to cool the electromagnet and the rheostat. The current I_{mag} can be varied by adjusting the rheostat. The lamps provide additional resistance and can be included or completely removed from the circuit.

How does reversing the orientation of the probe inside the gap affect the observed Hall voltage? What about switching the leads to the electromagnet? Are the polarities and directions consistent with those observed using the reference magnet?

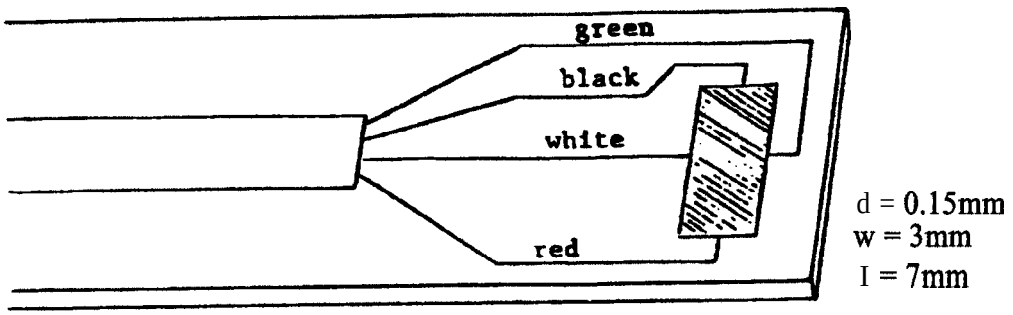


Figure 6

Part II - Cenco Probe

5. Plug in the circuit box for the Cenco probe (Figure 6) and set the current I to 150 mA. Place the Cenco probe in the pole gap of the electromagnet and determine V_H corresponding the magnetic field for a chosen current (e.g. I_{mag} = 5 amps). Repeat for several different values of I,,. Determine R_H and n for the Cenco probe. Which of the two probes is more sensitive?

6. From the direction of I, B_{mag}, polarity of V_H, and the orientation of the Cenco probe, determine the sign (+ or -) of the majority carriers for indium arsenide. Referring to Figure 6, record the potential differences (and polarity) between the various colored leads: red to black, white to black, green to black, and white to green. Carefully show all the relevant quantities in a labelled diagram and justify your answer for the type of charge carrier. (Hint: Consider all possible cases involving +q, -q, and direction of v .)

Questions

1. Why does the observed Hall voltage V_H vary as the probe is rotated in a magnetic field?
2. Is it better to have a high concentration or low concentration of charge carriers n when measuring very small magnetic fields? Why?
3. For metallic conductors with one valence electron per atom, the value for n is approximately equal to the number of atoms per unit volume. The metal silver is typical of such a behavior. Calculate n and R_H for silver using its atomic weight, density, and Avogadro's Number as follows:

Atomic weight A = 108gm; Density ρ = 10.5 gm/cm^3
Avogadro's Number N_o = 6.02 x 10^{23} atoms.